

Interior and Exterior Polygon Angle Sums

Answer Key

High School Geometry

Quick Answers

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|---------|-------|-------|-------|
| 1. 720 | 4. 30 | 7. 12 | 10. — |
| 2. 2340 | 5. 93 | 8. 36 | |
| 3. 135 | 6. 15 | 9. 13 | |
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1. Interior Angle Sum Theorem: a convex n -gon splits into $n - 2$ triangles from one vertex, and each triangle contributes 180° . A hexagon has $n = 6$, so the sum is $(6 - 2)180^\circ = 4 \cdot 180^\circ$.

720°

2. Same theorem with $n = 15$: the polygon splits into $15 - 2 = 13$ triangles, so the sum is $(15 - 2)180^\circ = 13 \cdot 180^\circ$. 2340°

3. An octagon has $n = 8$, so its interior angles total $(8 - 2)180^\circ = 1080^\circ$. *Regular* means all eight interior angles are congruent, so divide the total equally: $1080^\circ \div 8$. 135°

4. The exterior angles of any convex polygon total 360° , and a regular 12-gon has 12 congruent exterior angles. Divide: $360^\circ \div 12$. 30°

Note that the number of sides never enters through the interior-sum formula here — the exterior total is the same 360° for every convex polygon.

5. A pentagon has $n = 5$, so its interior angles total $(5 - 2)180^\circ = 540^\circ$. Set the five angles equal to that total:

$$102 + 118 + 96 + 131 + x = 540$$

The four known angles add to 447, so $x = 540 - 447$. $x = 93^\circ$

6. Every convex polygon's exterior angles total 360° . A regular polygon with n sides has n congruent exterior angles, so

$$24n = 360 \implies n = 360 \div 24$$

$n = 15$ sides. (Check: a regular 15-gon's interior angle is $180^\circ - 24^\circ = 156^\circ$.)

7. Working from the interior angle sum: a regular n -gon has n congruent interior angles that together make $(n - 2)180^\circ$, so

$$(n - 2)180 = 150n$$

Expand: $180n - 360 = 150n$, so $30n = 360$ and $n = 12$. $n = 12$ sides. (Faster check: the exterior angle is $180^\circ - 150^\circ = 30^\circ$, and $360 \div 30$ gives the same 12.)

8. A quadrilateral has $n = 4$, so its interior angles total $(4 - 2)180^\circ = 360^\circ$. The ratio $1 : 2 : 3 : 4$ means the angles are $x, 2x, 3x, 4x$:

$$x + 2x + 3x + 4x = 360 \implies 10x = 360$$

$x = 36^\circ$ (so the four angles are $36^\circ, 72^\circ, 108^\circ$, and 144° , which do add to 360°).

9. Set the interior angle sum formula equal to the given total and solve for the side count:

$$(n - 2)180 = 1980 \implies n - 2 = 11$$

$n = 13$ sides. A sum that is not a multiple of 180° would mean no such convex polygon exists, so dividing first is also the existence check.

10. Manual review — open justification. A complete argument contains:

- At each of the n vertices the interior angle and its exterior angle form a linear pair, so they add to 180° . Across all vertices, interior plus exterior totals $180n$ degrees.
- The interior angles alone total $(n - 2)180 = 180n - 360$ degrees.
- Subtracting, the exterior angles total $180n - (180n - 360) = 360$ degrees. The $180n$ terms cancel, so the result does not depend on n — it is 360° for every convex polygon.
- Equivalent accepted argument: walking once around the polygon, the turn at each vertex is that vertex's exterior angle, and one full circuit turns through exactly one complete rotation.

Accept any argument that is general in n and shows the n -dependence cancelling; do not accept a single worked example (for instance only a pentagon).